

Internet Appendix for “Principal Portfolios”

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ABSTRACT

This Internet Appendix provides additional tables and figures supporting the main text.

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I. On Mean-Variance Optimization of Linear Strategies

A. Robust Mean-Variance Framework of Linear Strategies

Consider the following robust mean-variance objective function:

$$\max_L E(S'_t L R_{t+1}) \text{ subject to } \max_{S: \|S\| \leq 1} \text{Var}_t(S' L R_{t+1}) \leq 1. \quad (\text{IA1})$$

This objective is a robust in the sense that we require that the variance be bounded regardless of the signal S . The variance term can be written as

$$\max_{S: \|S\| \leq 1} \text{Var}_t(S' L R_{t+1}) = \max_{S: \|S\| \leq 1} S' L \Sigma_{R,t} L' S = \max_{S: \|S\| \leq 1} \|\Sigma_{R,t}^{1/2} L' S\|^2 = \|\Sigma_{R,t}^{1/2} L'\|^2. \quad (\text{IA2})$$

So, if the assets are normalized and uncorrelated such that $\Sigma_{R,t} = \text{Id}$, then the robust variance constraint is the same as our matrix constraint $\|L\| \leq 1$.

We also get a similar solution when $\Sigma_{R,t} = \sigma^2 \text{Id}$ for some $\sigma \in \mathbb{R}$, since, in this case, the portfolio constraint simply becomes $\|L\| \sigma \leq 1$. Hence, we scale the position accordingly (i.e., use $L = L^*/\sigma$, where L^* is the standard solution with $\sigma = 1$). Note that when $\Sigma_{R,t} = \sigma^2 \text{Id}$, an alternative constraint is to require that the risk of the linear strategy cannot be greater than the risk of the simple factor as in (12), which, again, is equivalent to $\|L\| \leq 1$.

We can also easily solve a version of the robust mean-variance problem for general $\Sigma_{R,t}$. Solving (12) directly is not convenient since it would lead us to impose $\|\Sigma_{R,t}^{1/2} L'\| \leq 1$, which is not consistent with choosing a constant position matrix L . (Recall that the idea of linear strategies is to have a constant L but rich portfolio dynamics, $S'_t L$, driven by the signals.) Instead, we consider the transformed “synthetic assets” with returns

$$\tilde{R}_{t+1} = \Sigma_{R,t}^{-1/2} R_{t+1}, \quad (\text{IA3})$$

where signals are given by

$$\tilde{S}_t = \Sigma_{R,t}^{-1/2} S_t \quad (\text{IA4})$$

and the corresponding prediction matrix is $\tilde{\Pi} = E(\tilde{R}_{t+1} \tilde{S}'_t) = E(\Sigma_{R,t}^{-1/2} \Pi \Sigma_{R,t}^{-1/2})$. For example, if the original assets are uncorrelated (i.e., a diagonal $\Sigma_{R,t}$), then creating these synthetic

assets simply means scaling the assets to have the same constant volatility (as we do in the empirical analysis of futures).

Next, consider the robust mean-variance problem for synthetic assets:

$$\max_{\tilde{L}} E\left(\tilde{S}'_t \tilde{L} \tilde{R}_{t+1}\right) \text{ subject to } \max_{\tilde{S}_t} \text{Var}_t(\tilde{S}'_t \tilde{L} \tilde{R}_{t+1}) = \|\tilde{L}\| \leq 1. \quad (\text{IA5})$$

We see that this is our standard problem, expressed in terms of the synthetic assets. In other words, all of our results apply for the robust mean-variance problem of the synthetic assets. For example, the optimal strategy for the synthetic assets is $\tilde{L} = (\tilde{\Pi}'\tilde{\Pi})^{-1/2}\tilde{\Pi}'$, using Proposition 3. Of course, this solution can be translated back to the original assets by noting that

$$L_t = \Sigma_{R,t}^{-1/2} \tilde{L} \Sigma_{R,t}^{-1/2}, \quad (\text{IA6})$$

which holds since we must have that $\tilde{S}'_t \tilde{L} \tilde{R}_{t+1} = S'_t \Sigma_R^{-1/2} \tilde{L} \Sigma_R^{-1/2} R_{t+1} = S'_t L_t R_{t+1}$.

B. Risk Aversion instead of Risk Constraint

We can also consider a robust mean-variance problem in which the investor has risk aversion γ rather than a risk constraint:

$$\max_L \left(E(S'_t L R_{t+1}) - \frac{\gamma}{2} \left[\max_{S: \|S\| \leq 1} \text{Var}_t(S' L R_{t+1}) \right] \right). \quad (\text{IA7})$$

We can rewrite the objective using (IA2) as

$$\max_L \left(E(S'_t L R_{t+1}) - \frac{\gamma}{2} \|\Sigma_{R,t}^{1/2} L\|^2 \right). \quad (\text{IA8})$$

Let us first solve this portfolio problem when $\Sigma_{R,t} = \text{Id}$. To find the solution, we start by finding the solution for each level of volatility:

$$L(c) = \arg \max_{L: \|L\|=c} E(S'_t L R_{t+1}) = c(\Pi' \Pi)^{-1/2} \Pi', \quad (\text{IA9})$$

where the last equality uses Proposition 3. Now we can solve the objective function (IA12) by maximizing over all possible volatilities, c :

$$\max_c \left(E(S'_t L(c) R_{t+1}) - \frac{\gamma}{2} c^2 \right) = \max_c \left(c \sum_{i=1}^N \bar{\lambda}_i - \frac{\gamma}{2} c^2 \right) = \frac{(\sum_{i=1}^N \bar{\lambda}_i)^2}{2\gamma}, \quad (\text{IA10})$$

where the optimum is achieved by $c = \frac{\sum_{i=1}^N \bar{\lambda}_i}{\gamma}$, implying that the optimal strategy is

$$L = \frac{\sum_{i=1}^N \bar{\lambda}_i}{\gamma} (\Pi' \Pi)^{-1/2} \Pi', \quad (\text{IA11})$$

We see that the optimal strategy is the same as in Proposition 3 except for a scaling factor. The scaling factor naturally decreases in γ , reflecting that a more risk-averse investor takes a smaller position. Similarly, the scaling factor increases in the sum of the singular values, since higher singular values imply stronger predictability, leading to a larger position.

Finally, consider the problem with a general variance-covariance matrix $\Sigma_{R,t}$. Rather than solving the objective function (IA12) (which could lead to a time-varying L), we consider the similar objective function for the synthetic assets:

$$\max_{\tilde{L}} \left(E(\tilde{S}'_t \tilde{L} \tilde{R}_{t+1}) - \frac{\gamma}{2} \left[\max_{\tilde{S}: \|\tilde{S}\| \leq 1} \text{Var}_t(\tilde{S}' \tilde{L} \tilde{R}_{t+1}) \right] \right) = \max_{\tilde{L}} \left(E(\tilde{S}'_t \tilde{L} \tilde{R}_{t+1}) - \frac{\gamma}{2} \|\tilde{L}\|^2 \right). \quad (\text{IA12})$$

Since the synthetic assets have a variance-covariance matrix equal to the identity, their optimal solution is

$$\tilde{L} = \frac{\sum_{i=1}^N \bar{\lambda}_i(\tilde{\Pi})}{\gamma} (\tilde{\Pi}' \tilde{\Pi})^{-1/2} \tilde{\Pi}', \quad (\text{IA13})$$

which can be translated back to the original assets using (IA6).

C. Relation to the Standard Mean-Variance Framework

We next consider a standard mean-variance objective function, that is, we do not use the worst-case variance, but instead assume that conditional return variance is constant over time, $\Sigma_{R,t} = \Sigma_R$. In this case, we naturally recover the standard Markowitz solution. To see

this, we use the notation $\Sigma_S = E(S_t S_t')$ as before and calculate

$$\begin{aligned}
\max_L E \left(E_t(S_t' L R_{t+1}) - \frac{\gamma}{2} \text{Var}_t(S_t' L R_{t+1}) \right) &= \max_L E \left(S_t' L R_{t+1} - \frac{\gamma}{2} S_t' L \Sigma_R L' S_t \right) \\
&= \max_L \left(\text{tr}(L \Pi) - \frac{\gamma}{2} \text{tr}(L \Sigma_R L \Sigma_S) \right) \quad (\text{IA14}) \\
&= \max_{\tilde{L}} \left(\text{tr}(\tilde{L} \tilde{\Pi}) - \frac{\gamma}{2} \text{tr}(\tilde{L}' \tilde{L}) \right) \\
&= \max_{\tilde{L}} \left(\text{tr}(\tilde{L} \tilde{\Pi}) - \frac{\gamma}{2} \|\tilde{L}\|_2^2 \right),
\end{aligned}$$

where we use the change of variable $\tilde{L} = \Sigma_S^{1/2} L \Sigma_R^{1/2}$ and $\tilde{\Pi} = \Sigma_R^{-1/2} \Pi \Sigma_S^{-1/2}$. So we see that this problem has the same form as our normal objective function, except that we have another matrix norm, namely, the Frobenius 2-norm, $\|\cdot\|_2$. Since we solve the problem for all p -norms in Proposition 11 (and in the proof of Propostion 3), we know that solution, which is very simple: $\tilde{L} = c \tilde{\Pi}'$, where c is a constant that depends on risk aversion γ . So the solution to the mean-variance problem using the original variables is

$$L = \Sigma_S^{-1/2} \tilde{L} \Sigma_R^{-1/2} = c \Sigma_S^{-1/2} \tilde{\Pi}' \Sigma_R^{-1/2} = c \Sigma_S^{-1} \Pi \Sigma_R^{-1}. \quad (\text{IA15})$$

In other words, the optimal portfolio is

$$w_t = L' S_t = c \Sigma_R^{-1} \Pi \Sigma_S^{-1} S_t = c \Sigma_R^{-1} E(R_{t+1} | S_t), \quad (\text{IA16})$$

which is the standard Markowitz tangency portfolio (scaled by c depending on risk aversion). The last equality assumes that the conditional expected return can be computed using the multivariate regression of R_{t+1} on S_t , that is, $R_{t+1} = A S_t + \varepsilon_{t+1}$, and uses the fact that the regression coefficient is $A = \Pi \Sigma_S^{-1}$.

While our framework can nest the standard Markowitz solution, we seek to add robustness in several ways. First, we introduce the worst-case variance (captured by the operator matrix norm). Second, we avoid having to invert two matrices. Indeed, the Markowitz would first run a regression, requiring the matrix inversion Σ_S^{-1} , and then perform a portfolio optimization, requiring the matrix inversion, Σ_R^{-1} , which is known to be unstable in practice. Instead, we formulate a simpler objective function, leading to a solution that is simply a sum of singular vectors.

II. Proofs of Propositions

Proof of Proposition 1: Using the identity $\text{tr}(AB) = \text{tr}(BA)$ for any two square matrices A, B , we get

$$\text{tr}(E(S'_t L R_{t+1})) = \text{tr}(E(L R_{t+1} S'_t)) = \text{tr}(L E(R_{t+1} S'_t)) = \text{tr}(L \Pi) = \text{tr}(\Pi L).$$

□

Proof of Proposition 2: The proof follows directly from Proposition 1 and the fact that $\text{tr}(X'X) \geq 0$ for any matrix X . □

Proof of Proposition 3: We provide a proof in the case of a general Schatten p -norm considered in Proposition 11, where $p = [1, \infty]$ and q is defined by $1/p + 1/q = 1$. First, the trace of any square matrix $A = UDV'$ is less than the sum of its singular values (d_k),

$$|\text{tr}(A)| = |\text{tr}(UDV')| = |\text{tr}(V'UD)| = \left| \sum_k d_k (V'U)_{k,k} \right| \leq \sum_k d_k = \|A\|_1, \quad (\text{IA17})$$

since $|(V'U)_{k,k}| = |V'_{\text{column } k} U_{\text{column } k}| \leq \|V_{\text{column } k}\| \|U_{\text{column } k}\| = 1$. Combining this inequality with Hölder's inequality for Schatten norms (see, for example, Bhatia (1997), Corollary IV.2.6; or Tao (2012), p. 55, Exercise 1.3.9), we get:

$$|\text{tr}(L\Pi)| \leq \|\Pi L\|_1 \leq \|\Pi\|_q \|L\|_p. \quad (\text{IA18})$$

Finally, equality is achieved if L is proportional to $(\Pi'\Pi)^{q/2-1}\Pi'$. Thus,

$$\arg \max_{\|L\|_p \leq 1} \text{tr}(L\Pi) = (\Pi'\Pi)^{q/2-1}\Pi' / \|(\Pi'\Pi)^{q/2-1}\Pi'\|_p. \quad (\text{IA19})$$

□

Proof of Proposition 4: Follows from the calculations in the main text. □

Proof of Lemma 1: Suppose there exists a tradable factor $F_{t+1} = x'_t R_{t+1}$ such that $S_{i,t} = \frac{\text{Cov}_t(R_{i,t+1}, F_{t+1})}{\text{Var}_t(F_{t+1})}$. We then have

$$\text{Cov}_t(R_{i,t+1}, F_{t+1}) = \text{Cov}_t(R_{i,t+1}, x'_t R_{t+1}) = (\Sigma_t^R x_t)_i,$$

and hence $S_t = \Sigma_t^R x_t / y$, where we define $y = \text{Var}_t(F_{t+1})$. Furthermore,

$$\text{Var}_t(F_{t+1}) = x_t' \Sigma_t^R x_t.$$

Thus, we get $x_t = y(\Sigma_t^R)^{-1} S_t$. We therefore get a fixed point equation for y

$$y = \text{Var}_t(F_{t+1}) = x_t' \Sigma_t^R x_t = y^2 S_t' (\Sigma_t^R)^{-1} S_t \Leftrightarrow y = 1 / S_t' (\Sigma_t^R)^{-1} S_t.$$

Reverting the arguments, we see that the converse is also true: the just-computed portfolio x_t satisfies $S_{i,t} = \frac{\text{Cov}_t(R_{i,t+1}, F_{t+1})}{\text{Var}_t(F_{t+1})}$. $\square$

Proof of Proposition 5: By Lemma 2, we have

$$\begin{aligned} \text{tr}(L\Pi) &= \text{tr}((L^s + L^a)(\Pi^s + \Pi^a)) \\ &= \text{tr}(L^s \Pi^s) + \text{tr}(L^s \Pi^a) + \text{tr}(L^a \Pi^s) + \text{tr}(L^a \Pi^a) = \text{tr}(L^s \Pi^s) + \text{tr}(L^a \Pi^a). \end{aligned} \quad (\text{IA20})$$

It follows that

$$\begin{aligned} \text{Cov}_t(R_{t+1}^{w_t}, F_{t+1}) &= \text{Cov}_t(w_t' R_{t+1}, F_{t+1}) = w_t' \text{Cov}_t(R_{t+1}, F_{t+1}) \\ &= \text{Var}_t(F_{t+1}) w_t' S_t = \text{Var}_t(F_{t+1}) S_t' L S_t = \text{Var}_t(F_{t+1}) S_t' L^s S_t, \end{aligned} \quad (\text{IA21})$$

where the third identity uses the definition of F from Lemma 1 and the fact that $\text{Var}_t(F_{t+1}) = 1 / (S_t' (\Sigma_{R,t})^{-1} S_t)$, and the last identity follows because, by Lemma 2,

$$S_t' L S_t = S_t' L^s S_t + S_t' L^a S_t = S_t' L^s S_t. \quad (\text{IA22})$$

$\square$

Proof of Lemma 2: Since the trace of a matrix equals the trace of its transpose, we have

$$\text{tr}(AB) = \text{tr}((AB)') = \text{tr}(B'A') = -\text{tr}(BA) = -\text{tr}(AB),$$

which shows that $\text{tr}(AB) = \text{tr}(BA) = 0$. Similarly, $x' A x = (x' A x)' = x' A' x = -x' A x$, which implies that $x' A x = 0$. $\square$

Proof of Proposition 6: To see the first result, note that the return equation (25) combined

with the eigendecomposition (27) yields

$$E \left(S_t^{w_k^s} R_{t+1}^{w_k^s} \right) = (w_k^s)' \Pi^s w_k^s = (w_k^s)' \sum_{j=1}^K \lambda_j^s w_j^s (w_j^s)' w_k^s = \lambda_k^s. \quad (\text{IA23})$$

The last claim follows directly from (IA19) for Π^s because, for $L = L'$, by Proposition 5 we have that

$$\max_{\|L\|_p \leq 1, L=L'} \text{tr}(L\Pi) = \max_{\|L\|_p \leq 1, L=L'} \text{tr}(L\Pi^s)$$

is attained by the symmetric matrix $c((\Pi^s)' \Pi^s)^{q/2-1} (\Pi^s)' = (|\Pi^s|)^{q-2} \Pi^s$. For $q = 1$, we get $|\Pi^s|^{-1} \Pi^s = \text{sign}(\Pi^s)$. Here, we use the standard functional calculus for symmetric matrices (Horn and Johnson (1991)): for any function $f(x)$ (such as $|x|$ or $\text{sign}(x)$), we define $f(L) = W \text{diag}(f(\lambda(L))) W'$, where $L = W \text{diag}(\lambda(L)) W'$ is the eigendecomposition of a symmetric matrix L . $\square$

Proof of Proposition 7: Next, since $W = (w_1^s, \dots, w_N^s)$ forms an orthonormal basis of $\mathbb{R}^N$, we have $WW' = Id$, so

$$\tilde{F}_{t+1} = S_t' R_{t+1} = S_t' W W' R_{t+1} = (W' S_t) \cdot (W' R_{t+1}) = \sum_{k=1}^N S_t^{w_k^s} R_{t+1}^{w_k^s}. \quad (\text{IA24})$$

Hence, the result follows from the fact that including negative eigenvalues lowers the expected return relative to the other options considered. $\square$

Proof of Lemma 3: Equip $\mathbb{C}^N$ with the standard inner product

$$x \hat{\cdot} y = \sum_i x_i \bar{y}_i \quad (\text{IA25})$$

and recall that the Hermitian adjoint of a matrix B is defined as $B^* = \bar{B}'$, where $\bar{B}$ is the complex adjoint of B . Furthermore, for any matrix B , we have

$$x \hat{\cdot} (By) = (B^* x) \hat{\cdot} y. \quad (\text{IA26})$$

Let A now be a real antisymmetric matrix. Consider the matrix iA . The first observation is that iA is a Hermitian matrix. Indeed, $\overline{iA}' = -iA' = iA$. Thus, iA has real eigenvalues

and a basis of complex eigenvectors $\{w_k\}_{k=1}^N$. Let $\lambda \in \mathbb{R}$ be an eigenvalue of iA

$$iAw = \lambda w. \quad (\text{IA27})$$

Then taking a complex conjugate of this identity, we get

$$-iA\bar{w} = \lambda \bar{w}, \quad (\text{IA28})$$

and hence $\bar{w}$ is an eigenvector of iA with the eigenvalue $-\lambda$. Hence, all nonzero eigenvalues come in pairs.

Furthermore, $\det(A) = \det(A') = \det(-A) = (-1)^N \det(A)$, so, if N is odd, A is degenerate and has a zero eigenvalue, whereas all nonzero eigenvalues come in pairs. Let us take all nonnegative eigenvalues $\lambda_1 \geq \dots \geq \lambda_K$ of iA and let w_k be the respective complex eigenvectors. By the above, $-\lambda_k$ is also an eigenvalue, and the respective eigenvectors are $\bar{w}_k$. By the spectral decomposition theorem, we have

$$iA = \sum_{k=1}^K (\lambda_k w_k \bar{w}_k' - \lambda_k \bar{w}_k w_k'), \quad (\text{IA29})$$

where we use the fact that the orthogonal projection onto w_k is $w_k \bar{w}_k'$, with $\bar{w}_k$ the complex conjugate vector. We then have

$$\begin{aligned} \lambda_k w_k \bar{w}_k' - \lambda_k \bar{w}_k w_k' &= 0.5 \lambda_k (w_{k,1} + iw_{k,2})(w_{k,1} - iw_{k,2})' \\ &\quad - 0.5 \lambda_k (w_{k,1} - iw_{k,2})(w_{k,1} + iw_{k,2})' \\ &= i \lambda_k (w_{k,2} w_{k,1}' - w_{k,1} w_{k,2}') \end{aligned} \quad (\text{IA30})$$

and the claim follows.

For any Hermitian matrix (and hence for iA), eigenvectors for different eigenvalues are always orthogonal. Thus, $w_{k,1} \pm iw_{k,2}$ must be orthogonal to $w_{j,1} \pm iw_{j,2}$ and hence $w_{k,1}, w_{k,2}$ are orthogonal to $w_{j,1}, w_{j,2}$. Furthermore, $w_{k,1} \pm iw_{k,2}$ correspond to different eigenvalues $\pm \lambda_k$ and hence they also must be orthogonal,

$$0 = (w_{k,1} + iw_{k,2}) \cdot (w_{k,1} + iw_{k,2}) = \|w_{k,1}\|^2 - \|w_{k,2}\|^2 + 2iw_{k,1} \cdot w_{k,2}, \quad (\text{IA31})$$

which implies $\|w_{k,1}\| = \|w_{k,2}\|$ and $w_{k,1} \cdot w_{k,2} = 0$. Thus, the two vectors are $w_{k,1}, w_{k,2}$ are also orthogonal.

Note that

$$iA(w_{k,1} + iw_{k,2}) = \lambda_k(w_{k,1} + iw_{k,2}) \quad (\text{IA32})$$

is equivalent to $Aw_{k,1} = \lambda_k w_{k,2}$ and $Aw_{k,2} = -\lambda_k w_{k,1}$, which implies that, in the basis $\{(w_{k,1}, w_{k,2})\}_{k=1}^K$, the matrix A is block-diagonal, composed of diagonal blocks

$$\begin{pmatrix} 0 & -\lambda_1 \\ \lambda_1 & 0 \end{pmatrix}. \quad (\text{IA33})$$

□

Proof of Proposition 8: By definition, $(\Pi^a)'(x_j + iy_j) = i\lambda_j^a(x_j + iy_j)$, that is, $(\Pi^a)'x_j = -\lambda_j^a y_j$, $(\Pi^a)'y_j = \lambda_j^a x_j$. The expected return is

$$\begin{aligned} E(R_{t+1}^{w_{j,t}^a}) &= E(S_t'(x_j y_j' - y_j x_j') R_{t+1}) = \text{tr}((x_j y_j' - y_j x_j') \Pi^a) = -\text{tr}((\Pi^a)'(x_j y_j' - y_j x_j')) \\ &= \lambda_j^a \text{tr}(y_j' y_j + x_j' x_j) = 2\lambda_j^a. \end{aligned}$$

The last statement follows from (IA19) for Π^a because, for $L = -L'$, by Proposition 5 we have that

$$\max_{\|L\|_p \leq 1, L = -L'} \text{tr}(L\Pi) = \max_{\|L\|_p \leq 1, L = -L'} \text{tr}(L\Pi^a)$$

is attained by the antisymmetric matrix $c((\Pi^a)'\Pi^a)^{q/2-1}(\Pi^a)' = (|i\Pi^a|)^{q-2}(\Pi^a)'$. For $q = 1$, we get $|i\Pi^a|^{-1}(\Pi^a)' = \text{sign}(i\Pi^a)$. Here, we use the standard functional calculus for Hermitian matrices (Horn and Johnson (1991)): for any function $f(x)$ (such as $|x|$ or $\text{sign}(x)$), we define $f(L) = W \text{diag}(f(\lambda(L)))W'$, where $L = W \text{diag}(\lambda(L))W'$ is the eigen-decomposition of a Hermitian matrix L . □

Proof of Proposition 9: Follows from calculations in the body of the paper. □

Proof of Proposition 10: Direct calculation shows that

$$\Pi = (E[(\kappa_t + \beta_t)E_t[R_{i,t+1}]E_t[R_{j,t+1}]])_{i,j=1}^N,$$

which yields the result. □

Proof of Proposition 11: Let $X = L\Pi$. Then by a result of [Fan and Hoffman \(1955\)](#), we have $\lambda_i(X^s) \leq \bar{\lambda}_i(X)$. Furthermore, since $\text{rank}(L) \leq K$, we also have $\text{rank}(X) \leq K$ and $\text{rank}(X'X) = \text{rank}(X) \leq K$, and hence there are at most K nonzero singular values of X . Thus,

$$\text{tr}(L\Pi) = \text{tr}(X) = \text{tr}(X^s) = \sum_{i=1}^N \lambda_i(X^s) \leq \sum_{i=1}^K \bar{\lambda}_i(X).$$

Second, by a known result about singular values of products of matrices (see, for example, [Marshall and Olkin \(1979\)](#), p. 248), we have

$$\sum_{i=1}^K \bar{\lambda}_i(L\Pi) \leq \sum_{i=1}^K \bar{\lambda}_i(L) \bar{\lambda}_i(\Pi).$$

Third, by the Hölder inequality,

$$\sum_{i=1}^K \bar{\lambda}_i(L) \bar{\lambda}_i(\Pi) \leq \left(\sum_{i=1}^K \bar{\lambda}_i(L)^p \right)^{1/p} \left(\sum_{i=1}^K \bar{\lambda}_i(\Pi)^q \right)^{1/q} = \|L\|_p \left(\sum_{i=1}^K \bar{\lambda}_i(\Pi)^q \right)^{1/q}.$$

Thus,

$$\max_{\|L\|_p \leq 1, \text{rank}(L) \leq K} \text{tr}(L\Pi) \leq \left(\sum_{i=1}^K \bar{\lambda}_i(\Pi)^q \right)^{1/q}.$$

It remains to verify that the equality holds with $L = c \sum_{k=1}^K \bar{\lambda}_k^{q-1} v_k(u_k)'$, where $c = \left(\sum_{k=1}^K \bar{\lambda}_k^q \right)^{-1/p}$. This follows directly from the identity $\text{tr}(Lv_k(u_k)')$, established in [\(17\)](#)

Items 2 and 3 of the proposition are in fact special cases of item 1. Indeed, for item 2, we have by Proposition [5](#) that

$$\begin{aligned} \max_{\|L\|_p \leq 1, L=L', \text{rank}(L) \leq K} \text{tr}(L\Pi) &= \max_{\|L\|_p \leq 1, L=L', \text{rank}(L) \leq K} \text{tr}(L\Pi^s) \\ &\leq \max_{\|L\|_p \leq 1, \text{rank}(L) \leq K} \text{tr}(L\Pi^s) = \left(\sum_{i=1}^K \bar{\lambda}_i(\Pi^s)^q \right)^{1/q}. \end{aligned} \tag{IA34}$$

Furthermore, $\bar{\lambda}_i(\Pi^s)$ is the i -th largest absolute eigenvalue of Π^s and the equality is achieved with $L = c \sum_{k \in \mathcal{K}} |\lambda_k^s|^{q-1} \text{sign}(\lambda_k^s) w_k^s (w_k^s)'$

Similarly, for the antisymmetric part, we have

$$\begin{aligned}
\max_{\|L\|_p \leq 1, L = -L', \text{rank}(L) \leq K} \text{tr}(L\Pi) &= \max_{\|L\|_p \leq 1, L = -L', \text{rank}(L) \leq K} \text{tr}(L\Pi^a) \\
&\leq \max_{\|L\|_p \leq 1, \text{rank}(L) \leq K} \text{tr}(L\Pi^a) = \left(\sum_{i=1}^K \bar{\lambda}_i(\Pi^a)^q \right)^{1/q}.
\end{aligned} \tag{IA35}$$

The top $2K$ singular values of Π^a are just λ_i^a counted twice, and the equality is achieved with $L = c \sum_{k=1}^K (\lambda_k^a)^{q-1} (w_k^a (\bar{w}_k^a)' - \bar{w}_k^a (w_k^a)')$, where $w_k^a = x_k + iy_k$ are the *complex* eigenvectors of Π^a and come in complex conjugate pairs according to Lemma 3. $\square$

III. Additional Asset Pricing Tests (Extensions of Section III)

PROPOSITION IA1: *For any adapted stochastic process κ_t , let $\Pi(\kappa) = (E(\kappa_t R_{i,t+1} S_{j,t})_{i,j=1}^N) = E(\kappa_t R_{t+1} S_t')$. The following is true:*

- *We have $E_t(R_{i,t+1}) = \theta_t S_{i,t}$ with $\theta_t \geq 0$ if and only if $\Pi(\kappa)$ is symmetric and positive semi-definite for any adapted stochastic process $\kappa_t \geq 0$.*
- *We have $E_t(R_{i,t+1}) = \theta_t S_{i,t}$ with some θ_t if and only if $\Pi(\kappa)$ is symmetric for any adapted stochastic process κ_t .*

Proof of Proposition IA1: The “only if” part follows directly from calculations in the body of the paper. To prove the converse, we note first that, by the law of iterated expectations, $\Pi(\kappa)$ is symmetric if and only if

$$E(\kappa_t (E_t(R_{i,t+1}) S_{j,t} - E_t(R_{j,t+1}) S_{i,t})) = 0.$$

Since κ_t is arbitrary nonnegative, and any random variable can be written as a difference of two nonnegatives, we get that

$$E_t(R_{i,t+1}) S_{j,t} - E_t(R_{j,t+1}) S_{i,t} = 0.$$

This means that the vectors $E_t(R_{t+1})$ and S_t are co-linear and hence $E_t(R_{t+1}) = \theta_t S_t$ for some θ_t . Thus,

$$\Pi(\kappa) = E(\kappa_t \theta_t S_t S_t').$$

Suppose for simplicity that $S_t \neq 0$ with probability one. Then there exists a vector π such that $\pi' S_t \neq 0$ with probability one. It follows that $\pi' \Pi(\kappa) \pi = E(\kappa_t \theta_t (\pi' S_t)^2) > 0$ for all $\kappa_t \geq 0$, which implies $\theta_t \geq 0$. $\square$

PROPOSITION IA2 (Alpha and negative eigenvalues): *Suppose that*

$$R_{j,t+1} = \alpha_t + S_{j,t} F_{t+1} + \varepsilon_{j,t+1}, \quad E_t(\varepsilon_{j,t+1}) = 0,$$

as in [Frazzini and Pedersen \(2014\)](#). Then Π^s has at most one negative eigenvalue, while Π^a has at most one positive eigenvalue.

Furthermore, let $Q = (E(S_{j,t} \alpha_t))_{j=1}^N$, and define $\pi = Q/\|Q\| - \mathbf{1}_N N^{-1/2}$.

- $\Pi^a = 0.5(\mathbf{1}Q' - Q\mathbf{1}')$ is nonzero if and only if Q is not proportional to $\mathbf{1}_N$.
- If

$$E((\pi' S_t)^2 E_t(F_{t+1})) < (1 - N^{-1/2} Q' \mathbf{1} / \|Q\|)^2,$$

then Π^s has exactly one negative eigenvalue.²⁴

Proof of Proposition IA2: The proof of the first claim follows directly from Weyl inequalities ([IA50](#)) in complete analogy with the proof of Proposition [IA9](#). The second claim follows because

$$\pi' \Pi^s \pi = E((\pi' S_t)^2 E_t(F_{t+1})) - (1 - N^{-1/2} Q' \mathbf{1} / \|Q\|)^2,$$

and the inequality $\pi' \Pi^s \pi$ implies that Π^s has a negative eigenvalue. $\square$

PROPOSITION IA3 (Omitted factors and negative eigenvalues): *Suppose that*

$$R_{j,t+1} = S_{j,t} F_{t+1} + \beta_{j,t} F_{t+1}^1 + \varepsilon_{j,t+1}, \quad E_t(\varepsilon_{j,t+1}) = 0.$$

Suppose further that $\text{Cov}(S_{i,t} F_{t+1}^1, \beta_{j,t}) = 0$ for all i, j . Then Π^s has at most one negative eigenvalue, while Π^a has at most one positive eigenvalue. Moreover, let $\bar{S} = (E(S_{j,t} F_{t+1}^1))_{j=1}^N$, and $\bar{\beta} = (E(\beta_{j,t}))_{j=1}^N$, and define $\pi = \bar{S}/\|\bar{S}\| - \bar{\beta}/\|\bar{\beta}\|$. Then,

- $\Pi^a = 0.5(\bar{\beta} \bar{S}' - \bar{S} \bar{\beta}')$ is nonzero if and only if $\bar{S}$ is not proportional to $\bar{\beta}$.

²⁴This condition is only sufficient, but simple. It is possible to derive a necessary and sufficient condition, but it is more involved.

- If

$$E((\pi' S_t)^2 E_t(F_{t+1})) < (1 - \bar{S}' \bar{\beta})^2,$$

then Π^s has exactly one negative eigenvalue.²⁵

- The result extends to the case in which returns have a multi-factor structure. In which case Π^s has at most K negative eigenvalues if there are K omitted factors under the condition $\text{Cov}(S_{i,t} F_{t+1}^k, \beta_{j,k,t}) = 0$ for all i, j and all factors $k = 1, \dots, K$. Similarly,

$$\Pi^a = 0.5 \sum_k (\bar{\beta}_k \bar{S}'_k - \bar{S}_k \bar{\beta}'_k),$$

$$\text{where } \bar{S}_k = (E(S_{j,t} F_{t+1}^k))_{j=1}^N, \bar{\beta}_k = (E(\beta_{j,k,t}))_{j=1}^N.$$

Proof of Proposition IA3: The proof is completely analogous to that of Proposition IA2. □

PROPOSITION IA4 (Sign-changing risk premium or omitted factors): *Suppose that returns have a $(K + 1)$ -factor structure with some latent factor exposures $\beta_{i,k,t}$ such that*

$$E_t[R_{i,t+1}] = \theta_t S_{i,t} + \sum_{k=1}^K \gamma_{k,t} \beta_{i,k,t}.$$

Suppose also that

$$E[\kappa_t S_{j,t} \sum_{k=1}^K \gamma_{k,t} \beta_{i,k,t}] = 0$$

for all $i \neq j$. That is, there is no co-movement between factor exposures. Then:

- $\Pi(\kappa)$ is symmetric
- $\theta_t \geq 0$ if and only if $\Pi(\kappa)$ is positive semi-definite for any $\kappa_t \geq 0$.

²⁵This condition is only sufficient. It is possible to derive a necessary and sufficient condition, but it is more cumbersome.

IV. Central Limit Theorems for Asset Pricing Tests

We next develop central limit theorems (CLTs) for empirically testing the asset pricing bounds of Proposition 9. The starting point is to estimate of the prediction matrix, Π , by its sample counterpart:

$$\hat{\Pi}_{i,j}(T) = \frac{1}{T} \sum_{t=1}^T R_{i,t+1} S_{j,t}. \quad (\text{IA36})$$

Based on this estimate, the question is whether we can reject that the estimated $\hat{\Pi}$ satisfies the conditions in Proposition 9.

Proposition 9 rules out two types of alpha when signals are betas: one coming from negative eigenvalues of $\hat{\Pi}^s$ and another coming from a nonzero $\hat{\Pi}^a$. Hence, we test (i) whether the negative eigenvalues of $\hat{\Pi}^s$ are statistically significant or are just sampling noise, and (ii) whether $\hat{\Pi}^a$ is significantly different from zero. We use the following distributional assumption.

ASSUMPTION IA1: *Excess returns can be written as*

$$R_{i,t+1} = \theta_t S_{i,t} + \varepsilon_{i,t+1}, \quad (\text{IA37})$$

where ε_{t+1} are i.i.d. over time with zero mean and covariance matrix Σ^ε ; $\theta_t \geq 0$ are i.i.d. over time with finite first and second moments; signals $S_t \sim N(\bar{S}, \Sigma^S)$ are i.i.d. over time; S_t , θ_t , and ε_{t+1} are jointly independent; and the eigenvalues of Π^s are pairwise disjoint, $\lambda_1^s > \dots > \lambda_N^s$.

This assumption has several parts. First, (IA37) is simply a way of stating that we are working under the null hypothesis (44) from Proposition 9. Second, we make some distributional assumptions, and, lastly, we assume that none of the eigenvalues are the same, which holds generically (so this is a minimal loss of generality). The following proposition shows how to implement the first test.

PROPOSITION IA5 (Eigenvalue CLT): *Under Assumption IA1, in the limit as $T \rightarrow \infty$, the estimated eigenvalues $\hat{\Lambda}^s = (\hat{\lambda}_k^s)_{k=1}^N$ of $\hat{\Pi}^s$ converge almost surely to the true eigenvalues, $\Lambda^s = (\lambda_k^s)_{k=1}^N$, and $\sqrt{T}(\hat{\Lambda}^s - \Lambda^s)$ converges in distribution to $N(0, \Sigma^\Lambda)$, where Σ^Λ has elements*

$$\Sigma_{k,l}^\Lambda = \lambda_k^s \lambda_l^s \frac{\text{Var}(\theta)}{(E(\theta))^2} - 2E(\theta^2)(\bar{S}'w_k^s)^2(\bar{S}'w_l^s)^2 + \delta_{k,l}\lambda_k^s \left(2\frac{E(\theta^2)}{(E(\theta))^2}\lambda_k^s + \frac{(w_k^s)'\Sigma_\varepsilon w_k^s}{E(\theta)} \right). \quad (\text{IA38})$$

It is interesting that we can compute an asymptotic distribution of the ranked eigenvalues. The first two terms of (IA38) show the covariance between eigenvalues, while variances are driven by the last term since the kronecker delta, $\delta_{k,l}$, is zero when $k \neq l$ and one otherwise.²⁶

This proposition shows how to test Proposition 9. Suppose, for example, that the estimated prediction matrix $\hat{\Pi}^s$ has three negative eigenvalues collected in a vector $\lambda_{neg} = (\lambda_k^s)_{k=N-2}^N$. Then we can use (IA38) to compute their 3-by-3 variance-covariance matrix, $\Sigma_{neg} = (\Sigma_{k,l}^\Lambda)_{k,l \geq N-2}$, and the test whether these are in fact different from zero via $\lambda_{neg}'\Sigma_{neg}^{-1}\lambda_{neg}$, a Chi-squared test with three degrees of freedom.

Turning to the second test, we need to consider whether $\hat{\Pi}^a$ is significantly different from zero. Since $\hat{\Pi}^a$ is antisymmetric by construction, we know that the diagonal is zero and the lower triangle equals minus the upper triangle—so we only need to test that the upper triangle is zero. Accordingly, we focus on $Z_T = \sqrt{T} \text{vec} \left(\{\hat{\Pi}_{T,i,j}^a\}_{i=1,\dots,N,j>i} \right) \in \mathbb{R}^{\frac{N(N-1)}{2}}$, which is the upper triangle “stacked” into a vector (the vec-operator means stacking) and scaled by sample size, T , to ensure a nontrivial asymptotic distribution.

PROPOSITION IA6 (Antisymmetric CLT): *Let $\bar{\Sigma}_{i,j}^S = E(S_{i,t}S_{j,t})$. The vector Z_T is asymptotically normal with zero mean and asymptotic covariance between $Z_{(i,j)}$ and $Z_{(k,l)}$ given by*

$$\Sigma_{(i,j),(k,l)}^Z = \frac{1}{4} \left(\bar{\Sigma}_{i,k}^S \Sigma_{j,l}^\varepsilon - \bar{\Sigma}_{j,k}^S \Sigma_{i,l}^\varepsilon - \bar{\Sigma}_{i,l}^S \Sigma_{j,k}^\varepsilon + \bar{\Sigma}_{j,l}^S \Sigma_{i,k}^\varepsilon \right).$$

Hence, $Z_T'(\Sigma^Z)^{-1}Z_T$ is asymptotically χ^2 -distributed with $\frac{N(N-1)}{2}$ degrees of freedom. Further, $Z_T'Z_T$ has an asymptotic distribution given by $\sum_{k=1}^{\frac{N(N-1)}{2}} \lambda_k^Z Y_k$, where Y_k are i.i.d. χ_1^2 variables and $(\lambda_k^Z)_{k=1,\dots,\frac{N(N-1)}{2}}$ are the eigenvalues of Σ^Z .

This proposition shows two ways to test whether the antisymmetric part, captured in Z , is zero. One test is a standard Chi-squared test. The second test is based on a slightly

²⁶The second term in (IA38) can make the eigenvalues' covariances negative, which intuitively is related to the fact that their sum is fixed (as the trace of the prediction matrix). Clearly, variances must be positive, which follows from $\lambda_k^s = (w_k^s)'\Pi^s w_k^s = E(\theta)((w_k^s)'\Sigma^s w_k^s + (w_k^s)'\bar{S}\bar{S}'w_k^s) \geq E(\theta)((w_k^s)'\bar{S}\bar{S}'w_k^s) = E(\theta)(\bar{S}'w_k^s)^2$, which implies in turn that the third term is greater than the second, $\frac{E(\theta^2)}{(E(\theta))^2}(\lambda_k^s)^2 \geq E(\theta^2)(\bar{S}'w_k^s)^4$.

more complicated distribution, a weighted sum of Chi-squares, but this test has the potential benefit that we need not invert the asymptotic covariance matrix. Rather, we just need to compute its eigenvalues.

PROPOSITION IA7: *Fix a portfolio w . The finite-sample estimate*

$$\sqrt{T} \left(\frac{1}{T} \sum_{t=1}^T S_t^w R_{t+1}^w - w^\top \Pi^s w \right) \sim N(0, \sigma_s^2(\pi)),$$

where

$$\sigma_s^2(\pi) = (3(w' \bar{\Sigma}^S w)^2 - 2(\bar{S}' w)^4) E(\theta^2) + (w' \bar{\Sigma} w) w' \Sigma^\varepsilon w - (E(\theta) w' \bar{\Sigma}^S w)^2.$$

PROPOSITION IA8: *Fix a pair of portfolios x, y . The finite-sample estimate of the mean return on the basic alpha portfolio*

$$\sqrt{T} \left(\frac{1}{T} \sum_{t=1}^T (S_t^x R_{t+1}^y - S_t^y R_{t+1}^x) \right) \sim N(0, \sigma_a^2(x, y)),$$

where

$$\sigma_a^2(x, y) = x' \bar{\Sigma}^S x y' \Sigma^\varepsilon y + y' \bar{\Sigma}^S y x' \Sigma^\varepsilon x - 2x' \bar{\Sigma}^S y x' \Sigma^\varepsilon y.$$

A. Proof of CLTs

Proof of Proposition IA5: The proof will be based on several technical lemmas.

LEMMA IA1: *Consider a sequence of i.i.d. random vectors $X_t \in \mathbb{R}^d$ with finite first two moments and a covariance matrix Σ^X , and let*

$$\bar{X}_T = \frac{1}{T} \sum_{t=1}^T X_t.$$

Let $F : \mathbb{R}^d \rightarrow \mathbb{R}^K$ be a continuous function that is continuously differentiable in a small neighbourhood of $E(X)$, and let $A = DF(E(X))$ be the Jacobian of F evaluated at $E(X)$. Then the following CLT holds for F

$$\sqrt{T}(F(\bar{X}_T) - F(E(X))) \rightarrow N(0, DF \Sigma^X DF')$$

in distribution.

Proof of Lemma IA1: Standard results imply that it suffices to prove convergence of characteristic functions

$$E[e^{\alpha' \sqrt{T}(F(\bar{X}_T) - F(E(X)))}]$$

to that of the Gaussian distribution. Using the Taylor approximation, we get

$$\sqrt{T}(F(\bar{X}_T) - F(E(X))) = DF\sqrt{T}(\bar{X}_T - E(X)) + \sqrt{T}o(\|\bar{X}_T - E(X)\|).$$

By the CLT, $\sqrt{T}(\bar{X}_T - E(X)) \rightarrow N(0, \Sigma^X)$ in probability and hence the error term $Y_T = \sqrt{T}o(\|\bar{X}_T - E(X)\|)$ converges to zero in probability, whereas $Z_T = DF\sqrt{T}(\bar{X}_T - E(X))$ converges to $N(0, DF\Sigma^X DF')$ in distribution. Pick $\varepsilon, \delta > 0$. Then $\text{Prob}(|Y_T| > \varepsilon) < \delta$ for all large T . Hence,

$$E[e^{i\alpha' \sqrt{T}(F(\bar{X}_T) - F(E(X)))}] = E[e^{i\alpha'(Z_T + Y_T)}],$$

and we know that $e^{i\alpha' Z_T}$ converges to the Gaussian characteristic function. Now,

$$\begin{aligned} |E[e^{i\alpha'(Z_T + Y_T)}] - E[e^{i\alpha' Z_T}]| &= |E[e^{i\alpha'(Z_T + Y_T)} - e^{i\alpha' Z_T}]| \\ &= |E[e^{i\alpha'(Z_T + Y_T)} - e^{i\alpha' Z_T}]| \leq E[|e^{i\alpha' Y_T} - 1|] \\ &\leq E[|e^{i\alpha' Y_T} - 1|(\mathbf{1}_{|Y| > \varepsilon} + \mathbf{1}_{|Y| \leq \varepsilon})] \leq C|\alpha|\varepsilon + 2\text{Prob}(|Y_T| > \varepsilon), \end{aligned} \tag{IA39}$$

where we use the fact that $|e^{i\alpha'(Z_T + Y_T)}| = |e^{i\alpha' Y_T}| = |e^{i\alpha' Z_T}| = 1$ and where C is a large enough constant that $|e^{i\alpha x} - 1| \leq C|\alpha||x|$. Thus, making ε, δ sufficiently small, we see that the characteristic function comes arbitrarily close to that of Z_T and the claim follows. $\square$

LEMMA IA2: Let $\Lambda(A)$ be the function mapping a symmetric $N \times N$ matrix into the vector of its eigenvalues, arranged in decreasing order. Let A be the matrix with disjoint eigenvalues $\lambda_1(A) > \dots > \lambda_N(A)$, and let w_k be the corresponding eigenvectors. Then Λ is continuously differentiable (in fact, real analytic) in a small neighbourhood of A and

$$\lambda_i(A + B) = \lambda_i(A) + w_i' B w_i + O(\|B\|^2).$$

Proof: Standard results imply that $\lambda_i(X)$ is a real analytic function in a small neighborhood of A . Hence, it suffices to show that the gradient of λ_i is given by the map $B \rightarrow w_i' B w_i$.

Indeed, differentiating the identity

$$w_k(t)'(A + tB)w_k(t) = \lambda_i(A + tB)$$

with respect to t , we get

$$\frac{d}{dt}w_k(t)'(A + tB)w_k(t) + w_k(t)'Bw_k(t) + w_k(t)'(A + tB)\frac{d}{dt}w_k(t) = \frac{d}{dt}\lambda_i(A + tB). \quad (\text{IA40})$$

But $w_k(t)'(A + tB) = \lambda_i w_k'(t)$ and $(A + tB)w_k(t) = \lambda_i w_k(t)$. Furthermore, differentiating the identity $w_k'(t)w_k(t) = 1$, we get that $\frac{d}{dt}w_k(t)'w_k(t) = 0$. Substituting all this into (IA40), we get

$$w_k(t)'Bw_k(t) = \frac{d}{dt}\lambda_i(A + tB),$$

which is what had to be proved. □

LEMMA IA3: *Let $Y_{i,j} = S_i R_j$. Then, the covariance matrix of X is given by*

$$\Sigma_{i,j,k,l}^X = \text{Cov}(X_{i,j}, X_{k,l}) = \bar{\Sigma}_{i,k} \Sigma_{j,l}^\varepsilon + \text{Var}(\theta) \bar{\Sigma}_{i,j}^S \bar{\Sigma}_{k,l}^S + E(\theta)^2 \left(\bar{\Sigma}_{i,k}^S \bar{\Sigma}_{j,l}^S + \bar{\Sigma}_{i,k}^S \bar{\Sigma}_{j,l}^S - 2\bar{S}_i \bar{S}_j \bar{S}_k \bar{S}_l \right).$$

Proof of Lemma IA3: We have

$$\begin{aligned} E(X_{i,j} X_{k,l}) &= E(S_{i,t} R_{j,t+1} S_{k,t} R_{l,t+1}) = E(S_{i,t} S_{k,t} E_t(R_{j,t+1} R_{l,t+1})) \\ &= E(S_{i,t} S_{k,t} (\theta_t^2 S_{j,t} S_{l,t} + E_t(\varepsilon_{j,t+1} \varepsilon_{l,t+1}))) = E(\theta^2) E(S_{i,t} S_{k,t} S_{j,t} S_{l,t}) + E(S_{i,t} S_{k,t}) E(\varepsilon_{j,t+1} \varepsilon_{l,t+1}), \end{aligned} \quad (\text{IA41})$$

whereas

$$E(X_{i,j}) E(X_{k,l}) = E(\theta)^2 E[S_i S_j] E[S_k S_l].$$

By assumption, S is Gaussian and therefore

$$\begin{aligned}
& E(S_{i_1} S_{i_2} S_{j_1} S_{j_2}) \\
&= E((S_{i_1} - \bar{S}_{i_1} + \bar{S}_{i_1})(S_{i_2} - \bar{S}_{i_2} + \bar{S}_{i_2})(S_{j_1} - \bar{S}_{j_1} + \bar{S}_{j_1})(S_{j_2} - \bar{S}_{j_2} + \bar{S}_{j_2})) \\
&= E((S_{i_1} - \bar{S}_{i_1})(S_{i_2} - \bar{S}_{i_2})(S_{j_1} - \bar{S}_{j_1})(S_{j_2} - \bar{S}_{j_2})) \\
&+ \bar{S}_{i_1} \bar{S}_{i_2} \Sigma_{j_1, j_2}^S + \bar{S}_{i_1} \bar{S}_{j_1} \Sigma_{i_2, j_2}^S + \bar{S}_{i_1} \bar{S}_{j_2} \Sigma_{i_2, j_1}^S + \bar{S}_{i_2} \bar{S}_{j_1} \Sigma_{i_1, j_2}^S + \bar{S}_{i_2} \bar{S}_{j_2} \Sigma_{i_1, j_1}^S + \bar{S}_{j_1} \bar{S}_{j_2} \Sigma_{i_1, i_2}^S + \bar{S}_{i_1} \bar{S}_{i_2} \bar{S}_{j_1} \bar{S}_{j_2} \\
&= \Sigma_{i_1, i_2}^S \Sigma_{j_1, j_2}^S + \Sigma_{i_1, j_2}^S \Sigma_{i_2, j_1}^S + \Sigma_{i_1, j_1}^S \Sigma_{i_2, j_2}^S \\
&+ \bar{S}_{i_1} \bar{S}_{i_2} \Sigma_{j_1, j_2}^S + \bar{S}_{i_1} \bar{S}_{j_1} \Sigma_{i_2, j_2}^S + \bar{S}_{i_1} \bar{S}_{j_2} \Sigma_{i_2, j_1}^S + \bar{S}_{i_2} \bar{S}_{j_1} \Sigma_{i_1, j_2}^S + \bar{S}_{i_2} \bar{S}_{j_2} \Sigma_{i_1, j_1}^S + \bar{S}_{j_1} \bar{S}_{j_2} \Sigma_{i_1, i_2}^S + \bar{S}_{i_1} \bar{S}_{i_2} \bar{S}_{j_1} \bar{S}_{j_2} \\
&= \bar{\Sigma}_{i_1, i_2}^S \bar{\Sigma}_{j_1, j_2}^S + \bar{\Sigma}_{i_1, j_2}^S \bar{\Sigma}_{i_2, j_1}^S + \bar{\Sigma}_{i_1, j_1}^S \bar{\Sigma}_{i_2, j_2}^S - 2\bar{S}_{i_1} \bar{S}_{i_2} \bar{S}_{j_1} \bar{S}_{j_2}.
\end{aligned} \tag{IA42}$$

Hence,

$$\begin{aligned}
\text{Cov}(X_{i,j}, X_{k,l}) &= E(S_{i,t} S_{k,t}) E(\varepsilon_{j,t+1} \varepsilon_{l,t+1}) \\
&+ E(\theta^2) \left(\bar{\Sigma}_{i,j}^S \bar{\Sigma}_{k,l}^S + \bar{\Sigma}_{i,k}^S \bar{\Sigma}_{j,l}^S + \bar{\Sigma}_{i,k}^S \bar{\Sigma}_{j,l}^S - 2\bar{S}_i \bar{S}_j \bar{S}_k \bar{S}_l \right) - E(\theta)^2 E(S_{i,t} S_{j,t}) E(S_{k,t} S_{l,t}) \\
&= \bar{\Sigma}_{i,k}^S \Sigma_{j,l}^\varepsilon + \text{Var}(\theta) \bar{\Sigma}_{i,j}^S \bar{\Sigma}_{k,l}^S + E(\theta)^2 \left(\bar{\Sigma}_{i,k}^S \bar{\Sigma}_{j,l}^S + \bar{\Sigma}_{i,k}^S \bar{\Sigma}_{j,l}^S - 2\bar{S}_i \bar{S}_j \bar{S}_k \bar{S}_l \right).
\end{aligned} \tag{IA43}$$

□

We are now ready to prove the result. Note that $\Pi = \Pi^s = E(\theta) \bar{\Sigma}^S$. Since, by assumption, all eigenvalues of Π^s are pairwise disjoint, Lemma IA2 implies that Lemma IA1 is applicable with F being the map from the matrix to its vector of eigenvalues. Thus, eigenvalues will be asymptotically normal, and we can use the CLT to compute the covariance of eigenvalues. By Lemma IA2,

$$\sqrt{T}(\hat{\lambda}_k^s - \lambda_k^s) = \sqrt{T}(\lambda_k(\hat{\Pi}^s) - \lambda_k(\Pi^s)) \approx \sqrt{T}(w_k^s)'(\hat{\Pi}^s - \Pi^s)w_k^s.$$

Let

$$d\Pi_{i,j} = \sqrt{T}(\hat{\Pi}_{i,j} - \Pi_{i,j}).$$

Then, $E(d\Pi) = 0$ and, by Lemma IA1, the covariance matrix is asymptotically given by

$$\begin{aligned}
& \text{Cov}(\sqrt{T}(\hat{\lambda}_k^s - \lambda_k^s), \sqrt{T}(\hat{\lambda}_k^s - \lambda_k^s)) \\
&= \text{Cov}((w_k^s)' d\Pi^s w_k^s, (w_l^s)' d\Pi^s w_l^s) = E((w_k^s)' d\Pi^s w_k^s (w_l^s)' d\Pi^s w_l^s) \\
&= 0.25 E\left(\sum_{i_1, i_2} \sum_{j_1, j_2} (d\Pi_{i_1, i_2} + d\Pi_{i_2, i_1})(d\Pi_{j_1, j_2} + d\Pi_{j_2, j_1}) w_{k, i_1} w_{k, i_2} w_{l, j_1} w_{l, j_2}\right) \\
&= 0.25 \sum_{i_1, i_2} \sum_{j_1, j_2} \left(\Sigma_{i_1, i_2, j_1, j_2}^X + \Sigma_{i_2, i_1, j_1, j_2}^X + \Sigma_{i_1, i_2, j_2, j_1}^X + \Sigma_{i_2, i_1, j_2, j_1}^X \right) w_{k, i_1} w_{k, i_2} w_{l, j_1} w_{l, j_2},
\end{aligned} \tag{IA44}$$

where Σ^X is given in Lemma IA3. We have

$$\Sigma_{i_1, i_2, j_1, j_2}^X = \bar{\Sigma}_{i_1, j_1} \Sigma_{i_2, j_2}^\varepsilon + \text{Var}(\theta) \bar{\Sigma}_{i_1, i_2}^S \bar{\Sigma}_{j_1, j_2}^S + E(\theta)^2 \left(\bar{\Sigma}_{i_1, j_1}^S \bar{\Sigma}_{i_2, j_2}^S + \bar{\Sigma}_{i_1, j_2}^S \bar{\Sigma}_{i_2, j_1}^S - 2 \bar{S}_{i_1} \bar{S}_{i_2} \bar{S}_{j_1} \bar{S}_{j_2} \right).$$

Thus, by direct calculation,

$$\begin{aligned}
& 0.25 \sum_{i_1, i_2} \sum_{j_1, j_2} \left(\Sigma_{i_1, i_2, j_1, j_2}^X + \Sigma_{i_2, i_1, j_1, j_2}^X + \Sigma_{i_1, i_2, j_2, j_1}^X + \Sigma_{i_2, i_1, j_2, j_1}^X \right) w_{k, i_1} w_{k, i_2} w_{l, j_1} w_{l, j_2} \\
&= (w_k^s)' \bar{\Sigma}^S w_l^s (w_k^s)' \bar{\Sigma}^\varepsilon w_l^s + \text{Var}(\theta) (w_k^s)' \bar{\Sigma}^S w_k^s (w_l^s)' \bar{\Sigma}^S w_l^s \\
&+ 2E(\theta)^2 ((w_k^s)' \bar{\Sigma}^S w_l^s)^2 - (\bar{S}' w_k^s)^2 (\bar{S}' w_l^s)^2.
\end{aligned} \tag{IA45}$$

Since eigenvectors are orthogonal, we have $(w_k^s)' \bar{\Sigma}^S w_l^s = \lambda_k^S \delta_{k, l}$, so the last expression is given by

$$\delta_{k, l} \lambda_k^S (w_k^s)' \bar{\Sigma}^\varepsilon w_k^s + \text{Var}(\theta) \lambda_k^S \lambda_l^S + 2E(\theta)^2 (\delta_{k, l} (\lambda_k^S)^2 - (\bar{S}' w_k^s)^2 (\bar{S}' w_l^s)^2),$$

and formula (IA38) follows because $\lambda_k^S = \lambda_k^s / E(\theta)$. $\square$

Proof of Proposition IA6: We have $E[S_{i, t} \varepsilon_{j, t+1} - S_{j, t} \varepsilon_{i, t+1}] = 0$ and hence

$$\begin{aligned}
& \text{Cov}((\Pi_{i, j} - \Pi_{j, i}), (\Pi_{k, l} - \Pi_{l, k})) = E((\Pi_{i, j} - \Pi_{j, i})(\Pi_{k, l} - \Pi_{l, k})) \\
&= \frac{1}{T^2} E\left(\left(\sum_{t=1}^T (S_{i, t} \varepsilon_{j, t+1} - S_{j, t} \varepsilon_{i, t+1}) \right) \left(\sum_{t=1}^T (S_{k, t} \varepsilon_{l, t+1} - S_{l, t} \varepsilon_{k, t+1}) \right) \right) \\
&= \frac{1}{T^2} T E((S_i \varepsilon_j - S_j \varepsilon_i)(S_k \varepsilon_l - S_l \varepsilon_k)) = \frac{1}{T} \left(\bar{\Sigma}_{i, k}^S \Sigma_{j, l}^\varepsilon - \bar{\Sigma}_{j, k}^S \Sigma_{i, l}^\varepsilon - \bar{\Sigma}_{i, l}^S \Sigma_{j, k}^\varepsilon + \bar{\Sigma}_{j, l}^S \Sigma_{i, k}^\varepsilon \right).
\end{aligned} \tag{IA46}$$

The claim now follows directly from the CLT for i.i.d. random variables. $\square$

V. Theoretical Relation to Haddad-Kozak-Santosh

Here, we put the analysis of [Haddad et al. \(2020\)](#) into the general framework of principal portfolios and show how principal portfolios can be used to (1) understand the origins and structure of predictability in [Haddad et al. \(2020\)](#) and (2) significantly improve the performance of factor-timing strategies from [Haddad et al. \(2020\)](#).

Let $\tilde{R} = R_{PC}$ and $\tilde{S} = S_{PC}$. Our first observation is that the prediction matrix $\Pi = (\tilde{R}'_{t+1}\tilde{S}_t)$ computed in sample is dominated by noise: The results of [Haddad et al. \(2020\)](#) imply that naively exploiting the full Π matrix can lead to poor predictive performance. As we know from formula (32),

$$E\left(\sum_{i=1}^{50} R_{t+1}^{q_i} S_t^{q_i}\right) = E\left(\sum_{i=1} \tilde{R}_{i,t+1} \tilde{S}_{i,t}\right) = \sum_i \Pi_{i,i} = \text{tr}(\Pi) = \text{tr}(\Pi^s),$$

where $\{q_i\}$ is the orthonormal basis of principal components. The results of [Haddad et al. \(2020\)](#) imply that even an estimate of the trace of the prediction matrix is noisy and does not work well out-of-sample. Furthermore, any attempt of a dimension reduction depends on the structure of eigenvalues of the Π matrix. Another key observation is that all trading strategies investigated in [Haddad et al. \(2020\)](#) fall into the class of symmetric linear strategies. For example, their factor-timing strategy satisfies

$$\tilde{R}_{t+1}^{\pi_t} = \pi_t' \tilde{R}_{t+1} = (\Sigma_{\tilde{R}}^{-1}(a + \text{diag}(b)S_t))' \tilde{R}_{t+1} = (\Sigma_{\tilde{R}}^{-1}a)' \tilde{R}_{t+1} + \tilde{S}_t' L \tilde{R}_{t+1},$$

where the matrix

$$L = \Sigma_{\tilde{R}}^{-1} \text{diag}(b)$$

is symmetric (because $\Sigma_{\tilde{R}}$, the in-sample covariance of in-sample PCs, is diagonal). Thus, the dynamic part of the factor-timing strategy attempts to exploit a low-rank linear symmetric strategy. Furthermore, empirical estimates of the b coefficients in [Haddad et al. \(2020\)](#) are all nonnegative, implying that L is in fact positive semi-definite. The following lemma complements Proposition 6 and provides an exact upper bound on the expected return of the factor-timing strategy.

LEMMA IA4: Let $\lambda_1^s \geq \dots \geq \lambda_N^s$ be the eigenvalues of the matrix Π^s and $\bar{\lambda}_1 \geq \dots \geq \bar{\lambda}_N$ the singular values of Π , let $(q_i)_{i=1}^N$ be an arbitrary orthonormal basis of $\mathbb{R}^N$ and $M \leq N$, let L be an arbitrary $M \times M$ symmetric positive semi-definite²⁷ symmetric matrix with eigenvalues $\lambda_1(L) \geq \dots \geq \lambda_M(L) \geq 0$, let $\hat{S}_t = (\tilde{S}_t^{q_i})_{i=1}^M$ and $\hat{R}_{t+1} = (\tilde{R}_{t+1}^{q_i})_{i=1}^M$, and let $\pi_t = L' \hat{S}_t$. Then

$$E\left(\hat{R}^{\pi_t}\right) \leq \sum_{i=1}^M \mathbf{1}_{\lambda_i^s > 0} E(PEP_i) \lambda_i(L) \leq \sum_{i=1}^M \lambda_i(L) E(PP_i). \quad (\text{IA47})$$

The first upper bound is attained if and only if $\lambda_i^s \geq 0$ for all $i \leq M$ and q_i is the respective eigenvector of Π^s . The second upper bound is attained if and only if, in addition, Π is symmetric and $\lambda_M^s \geq 0$.

The result of Lemma IA4 implies that no matter which rotation (using principal components or otherwise) we might use to exploit differences in predictability, we are limited by the eigenvalue structure of the Π^s matrix. Furthermore, if Π^s is not symmetric (i.e., $\Pi^a \neq 0$), then there will be a nonzero “gap” in the achievable excess return because the L for the factor-timing strategy is symmetric and hence cannot capture the antisymmetric part of predictability due to the identity (23). Of course, it may happen that the risk-return trade-off for some arbitrary factor-timing strategy appears better than principal portfolios. If so, it must be the case that the superior performance originates from better diversification, not from better return prediction.

A key insight of the principal portfolio analysis is that eigenvectors for the largest eigenvalues of the different parts of the Π matrix capture a robust part of predictability and hence tend to perform better than alternative prediction techniques out-of-sample.

VI. Cross-Sectionally Demeaning the Signals or the Returns

We start with a signal S , which is not cross-sectionally demeaned, and use the notation “ $\sim$ ” (tilde) to indicate demeaning:

$$\tilde{S}_{j,t} = S_{j,t} - \frac{1}{N} \sum_{k=1}^N S_{k,t}.$$

²⁷An analogous result holds when L is not positive semi-definite, in which case the upper bound is expressed in terms of the largest absolute values of the eigenvalues of Π^s .

Similarly, the prediction matrix based on demeaned signals is

$$\tilde{\Pi} = (R_{i,t+1} \tilde{S}_{j,t})_{i,j=1}^N,$$

and $\tilde{\lambda}_k^s$ and $\tilde{\lambda}_k^a$ are the eigenvalues of $\tilde{\Pi}^s$ and $\tilde{\Pi}^a$, respectively.

PROPOSITION IA9: *The demeaned eigenvalues $\tilde{\lambda}_k^s$ are interlacing with the nondemeaned ones λ_k^s in the sense that*

$$\lambda_{k+1}^s \leq \tilde{\lambda}_k^s \leq \lambda_{k-1}^s \quad (\text{IA48})$$

for all $k = 2, \dots, K-1$, $\lambda_2^s \leq \tilde{\lambda}_1^s$, and $\tilde{\lambda}_N^s \leq \lambda_{N-1}^s$, and similarly for the antisymmetric eigenvalues. Hence, if Π^s has N^p positive eigenvalues, then $\tilde{\Pi}^s$ has between N^p-1 and N^p+1 positive eigenvalues. Furthermore, the total performance of the cross-sectional factor

$$E \left(\sum_{i=1}^N \tilde{S}_{i,t} R_{i,t+1} \right) = \sum_{i=1}^N \tilde{\lambda}_i^s$$

satisfies

$$\sum_{i=2}^N \lambda_i^s \leq E \left(\sum_{i=1}^N \tilde{S}_{i,t} R_{i,t+1} \right) \leq \sum_{i=1}^{N-1} \lambda_i^s. \quad (\text{IA49})$$

In particular, if all eigenvalues λ_i^s are positive, then the cross-sectional factor performs worse than the time-series factor.

Proof of Proposition IA9: We use Weyl inequalities ([Horn and Johnson \(1991\)](#)): for any two symmetric or Hermitian matrices A , B ,

$$\lambda_j(A) + \lambda_k(B) \leq \lambda_i(A+B) \leq \lambda_r(A) + \lambda_s(B) \quad (\text{IA50})$$

whenever $j+k-N \geq i \geq r+s-1$.

Next, we note that, by direct calculation,

$$\tilde{\Pi}^s = \Pi^s + X,$$

where $X = 0.5(\pi \mathbf{1}' + \mathbf{1} \pi')$, and where the vector $\pi = (\pi_i) = -(E(R_{i,t+1} \frac{1}{N} \sum_{k=1}^N S_{k,t}))$. The matrix X has rank two and at most two nonzero eigenvalues that always have opposite

signs: $\lambda_1(X) \geq 0 \geq \lambda_N(X)$. Thus, by the Weyl inequalities,

$$\lambda_{i+1}(\Pi^s) \leq \lambda_N(X) + \lambda_{i+1}(\Pi^s) \leq \lambda_i(\tilde{\Pi}^s) \leq \lambda_{i-1}(\Pi^s) + \lambda_2(X) \leq \lambda_{i-1}(\Pi^s).$$

The proof for the antisymmetric part is analogous.

To prove the last inequality, define the orthogonal projection $P = Id - \frac{1}{N} \mathbf{1}_{N \times N}$. Then $\tilde{S}_t = PS_t$ and hence, by direct calculation, $\tilde{\Pi} = \Pi P$. Furthermore, since signals are demeaned,

$$\sum_{i=1}^N \tilde{S}_{i,t} R_{i,t+1} = \sum_{i=1}^N \tilde{S}_{i,t} \tilde{R}_{i,t+1},$$

where

$$\tilde{R}_{t+1} = PR_{t+1},$$

and hence

$$\begin{aligned} E \left(\sum_{i=1}^N \tilde{S}_{i,t} R_{i,t+1} \right) &= E \left(\sum_{i=1}^N \tilde{S}_{i,t} \tilde{R}_{i,t+1} \right) = \text{tr}(E(\tilde{R}_{t+1} \tilde{S}_t')) \\ &= \text{tr}(P\Pi P) = \text{tr}(P\Pi^s P). \end{aligned} \tag{IA51}$$

The eigenvalues of $P\Pi^s P$ coincide with the $N - 1$ eigenvalues $\{\hat{\lambda}_k^s\}_{k=1}^{N-1}$ of $P\Pi^s P|_{P\mathbb{R}^N}$ restricted onto the subspace $P\mathbb{R}^N$, plus a zero eigenvalue. By the interlacing inequalities [Horn and Johnson \(1991\)](#), we have $\lambda_{k+1}^s \leq \hat{\lambda}_k^s \leq \lambda_k^s$, and therefore

$$\text{tr}(P\Pi^s P) = \sum_{k=1}^{N-1} \hat{\lambda}_k^s$$

satisfies the required inequalities. □

The same arguments imply that the following is true.

PROPOSITION IA10: *Let*

$$\tilde{R}_{i,t+1} = R_{i,t+1} - \frac{1}{N} \sum_{k=1}^N R_{k,t+1}$$

and

$$\hat{\Pi} = (\tilde{R}_{i,t+1} S_{j,t})_{i,j=1}^N.$$

Then, the eigenvalues of $\hat{\Pi}^a$ and $\hat{\Pi}^s$ satisfy the same inequalities as those of $\tilde{\Pi}$ (see Proposition IA9).

A. Cross-Sectionally Demeaning Portfolio Signals

Given any orthonormal tuple of portfolios $\{\pi_k\}_{k=1}^K$, we define the corresponding demeaned signals as

$$\tilde{S}_t^{\pi_k} = S_t^{\pi_k} - \frac{1}{K} \sum_{i=1}^K S_t^{\pi_i}.$$

Interestingly, we “lose” one eigenvalue when using demeaned factors rather than non-demeaned ones as seen in the following proposition where $\lambda_1^s \geq \dots \geq \lambda_N^s$ are still the eigenvalues of Π^s (i.e., based on the original, non-demeaned signals).

PROPOSITION IA11: *The expected excess return of demeaned portfolios based on any orthonormal tuple of portfolios $\{\pi_k\}_{k=1}^K$ satisfies*

$$\sum_{i=1}^{K-1} \lambda_k^s \geq E\left(\sum_{k=1}^K \tilde{S}_t^{\pi_k} R_{t+1}^{\pi_k}\right) \geq \sum_{i=N-K+2}^N \lambda_k^s$$

and the bounds are exact.

Proof of Proposition IA11: Let

$$\tilde{\pi}_k = \pi_k - \frac{1}{K} \sum_{i=1}^K \pi_i.$$

Then,

$$\tilde{S}_t^{\pi_k} = S_t^{\tilde{\pi}_k},$$

and, since $\sum_k S_t^{\tilde{\pi}_k} = 0$, we have

$$\sum_k S_t^{\tilde{\pi}_k} R_{t+1}^{\pi_k} = \sum_k S_t^{\tilde{\pi}_k} R_{t+1}^{\tilde{\pi}_k}.$$

Define the matrix

$$X = (\tilde{\pi}_1, \dots, \tilde{\pi}_K),$$

with the columns given by $\tilde{\pi}_i$. Then,

$$E\left(\sum_k S_t^{\tilde{\pi}_k} R_{t+1}^{\tilde{\pi}_k}\right) = \text{tr}(X' \Pi^s X) = \text{tr}(\Pi^s X X').$$

Since, by assumption, π_k are orthonormal, we have

$$\tilde{\pi}_k' \tilde{\pi}_l = (\pi_k - \frac{1}{K} \sum_{i=1}^K \pi_i)' (\pi_l - \frac{1}{K} \sum_{i=1}^K \pi_i) = \delta_{k,l} - 1/K,$$

and hence the matrix $P = X'X \in \mathbb{R}^{K \times K}$ has rank $K - 1$ and eigenvalues 1 (of multiplicity $K - 1$) and 0, and hence is an orthogonal projection. Thus, we can write $X = UP$, where $U \in \mathbb{R}^{N \times K}$ is an orthogonal matrix satisfying $U'U = Id_K$. It follows that

$$\text{tr}(X' \Pi^s X) = \text{tr}(PU' \Pi^s UP).$$

Let V be the orthogonal matrix such that $\tilde{P} = V'PV$ is the projection onto the span of the first $K - 1$ standard basis vectors of $\mathbb{R}^N$. Then,

$$\text{tr}(PU' \Pi^s UP) = \text{tr}(PU' \Pi^s UPV V') = \text{tr}(V' P V V' U' \Pi^s U V V' P V) = \text{tr}(\tilde{P} \tilde{U}' \Pi^s \tilde{U} \tilde{P}),$$

where $\tilde{U} = UV$. Then, $\tilde{U}$ is an arbitrary orthogonal matrix with columns $u_1, \dots, u_k$, and

$$\text{tr}(\tilde{P} \tilde{U}' \Pi^s \tilde{U} \tilde{P}) = \sum_{i=1}^{K-1} u_i' \Pi^s u_i.$$

The claim then follows from the Ky Fan inequality ([Fan \(1950\)](#)). □

VII. Additional Empirical Results

A. Robustness of Momentum Results

This appendix reports extension and robustness analysis to complement the base-case portfolio momentum analysis of Section V.A. We study portfolio momentum via principal portfolios in several other data sets, including 25 U.S. size and operating profitability portfolios, 25 U.S. size and investment portfolios, the international counterparts of the three sets of Fama-French portfolios (i.e., developed countries excluding the U.S.), and a sample of 52 futures contracts starting in 1985. Futures data include contracts for 21 commodities, 17 equity indices, five sovereign bonds, and nine currencies.

While the base case signal is each asset’s 20-day momentum (approximately one month), we also consider momentum signals for robustness, namely, based on 40-, 60-, 90-, 120-, and 250-day past returns. In each case, we standardize the signal each period by converting it to a cross-sectional rank and mapping it into the $[-0.5, 0.5]$ interval.

Turning to returns, the base-case measure of returns is each asset’s 20-day return (again, one month). We also consider other forecast horizons, namely one-day, five-day, and 10-day returns, cross-sectionally demeaning returns to focus prediction on cross-sectional differences in returns. Similarly to the base case, each of the other forecast horizons corresponds to a sample of nonoverlapping time periods t of the same length. As advocated by Moskowitz et al. (2012), in the case of futures contracts we time-series devolatilize returns (both in the forecast target and in the momentum signal construction) using trailing 20-day return volatility, which helps avoid a situation in which results are unduly driven by the large cross-sectional differences in volatility of raw futures contract returns.²⁸

Figure IA1 shows that principal portfolios perform better at shorter forecast horizons, especially the principal portfolio and PEP strategies. Indeed, at the shorter forecast horizons, even PEP earns a higher Sharpe Ratio (SR) than the simple factor, and the risk-adjusted return as measured by the information ratio (IR) becomes highly significant at one-, five-, and 10-day forecast horizons.

Figure IA2.A and B show the SRs and IRs for seven other data sets: three samples of U.S. stocks (the base case from before, plus two other sets of Fama-French portfolios), three sets of international stocks (i.e., global stocks outside the U.S. sorted into similar portfolios), and a set of 52 futures contracts (consisting of equity index futures, bond futures, commodity

²⁸This adjustment has tiny effects in our equity asset analysis and thus, in the interest of simplicity, we do not de-volatilize equity returns.

futures, and currency forwards). In support of the model’s predictive power, we see that all of the SRs and IRs are positive, and several though not all are statistically significant. In further support of theoretical predictions, Figure IA2.C shows that the ex-ante eigenvalues are highly correlated to the ex-post realized returns in each sample.

Finally, we analyze the robustness of our method across momentum horizons and subsamples. Figure IA3 shows the performance of the leading principal portfolios, PEPs, and PAPs for different look-back periods in the specification of the momentum signal. Panel A shows that the PEP performs similarly to the simple factor for all momentum horizons. However, principal portfolio, PAP, and the PEP/PAP combination deliver higher SR across all horizons. Turning to the IRs in Panel B, we see that these are more statistically significant for the short- and medium-term momentum periods, and less significant for the longer-term momentum horizons.

Figure IA4 reports the performance of the base-case strategy for each decade in the sample. We see that the performance tends to be positive across decades — so the strong overall performance is not being driven by a single decade — but, naturally, the statistical significance in each decade is reduced due to the short time window.

Lastly, Figure IA5 shows that base-case results are qualitatively similar when we omit cross-sectional demeaning of signals and returns from our data construction and instead allow for timing effects to enter into principal portfolio estimates.

B. Testing the Fama-French Five-Factor Model

In this subsection, we illustrate how one could use results in [section III](#) and [section III–IV](#) for asset pricing tests. Our test universe consists of the 25 Fama-French portfolios sorted by value and book-to-market, with monthly returns $R_{i,t+1}$, $i = 1, \dots, 25$. We look at monthly data for the period July 1963 to January 2020. Our signals are constructed as follows. First, we compute $\beta_{i,t}(k)$, $k = 1, \dots, 5$ as rolling 24-month betas of Fama-French portfolio returns with respect to the Mkt-Rf, SMB, HML, RMW, and CMA, also from Kenneth French’s website. We then run a panel regression

$$R_{i,t+1} - r_t^F = \sum_{k=1}^5 \gamma_k \beta_{i,t}(k) + \varepsilon_{i,t+1}$$

over the full sample and define

$$S_{i,t} = \sum_{k=1}^5 \gamma_k \beta_{i,t}(k).$$

Intuitively, we expect that $S_{i,t}$ should be the “true” conditional expected return on $R_{i,t+1}$, and hence the Π matrix (estimated over the whole sample) should be symmetric and positive definite. To be able to apply Propositions [IA6](#) and [IA5](#), we need an estimate of $E(\theta)$, $\text{Var}(\theta)$, and Σ^ε . We

- Estimate θ_t by running a cross-sectional regression every month in the data sample, and then compute its sample mean and variance.^{[29](#)}
- Estimate the “true” $\Pi^s = E(\theta)\bar{\Sigma}^S$.
- Compute residuals $\varepsilon_{i,t+1} = R_{i,t+1} - \theta_t S_{i,t}$ and their sample covariance matrix Σ^ε .

For Propositions [IA7](#) and [IA8](#), we use, respectively, w as the PEP for the lowest negative eigenvalue of $\hat{\Pi}^s$, and x, y as the parts of the PAP for the largest positive eigenvalue of $\hat{\Pi}^a$.^{[30](#)}

We compute p -values as follows.

- For Proposition [IA5](#), the Chi-squared statistic for all negative eigenvalues is < 0.0001 so the null is rejected. The distribution of eigenvalues of Π^s is almost symmetric around zero if we exclude the largest positive eigenvalue. In particular, 12 out of 25 eigenvalues are negative.
- For Proposition [IA6](#), the matrix is not invertible. The generalized Chi-squared statistic gives a p -value of 0.3, so we cannot reject the null hypothesis. The test is too noisy and hence is not very helpful.
- For Proposition [IA7](#), p -value is 0.002 for the lowest (most negative) eigenvalue and 0.006 for the second lowest, so we reject the null that these eigenvalues are positive.
- For Proposition [IA8](#), p -value is < 0.0001 for the two largest antisymmetric eigenvalues, so we reject the null.

²⁹Our variance estimate is huge, massively inflating standard errors and hence making the null even harder to reject.

³⁰Note however, that, asymptotically, the realized variance of the respective portfolio returns in Propositions [IA7](#) and [IA8](#) coincides with the variance predicted by the CLT. Since these returns are asymptotically Gaussian, we can just use the respective SRs as the statistics.

Finally, we note that since $S_{i,t}$ was estimated so that $S_{i,t} \approx E_t[R_{i,t+1}]$, one might also argue that $\theta_t = 1$. We consider this case as well. The corresponding test statistics slightly improve because $\text{Var}(\theta) = 0$.

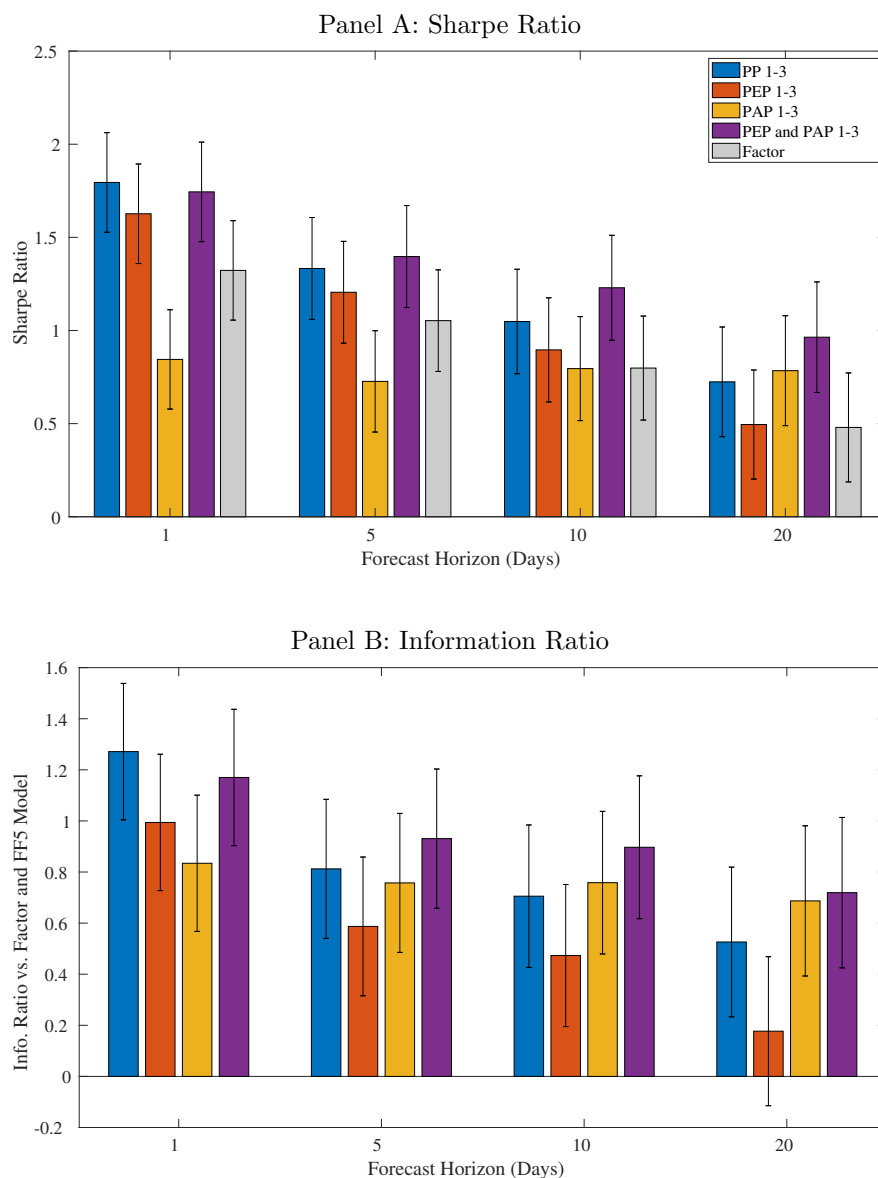
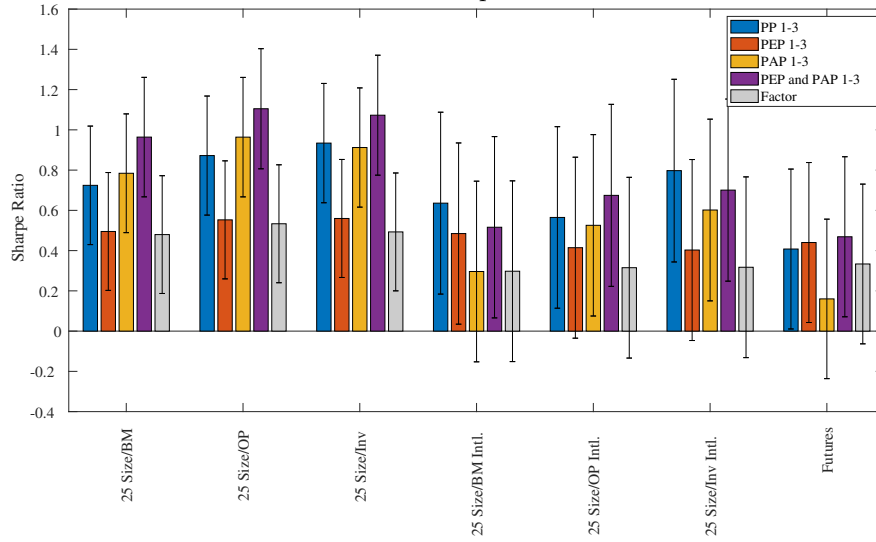
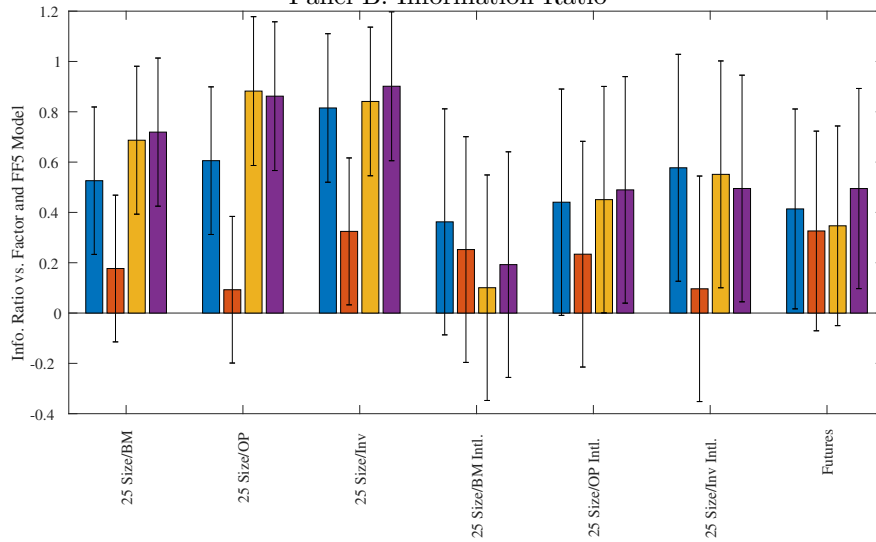


Figure IA1. Principal portfolio performance by forecast horizon. This figure shows out-of-sample performance of principal portfolios in terms of the annualized Sharpe ratio (Panel A) and the annualized information ratio versus the own-predictor strategy and the Fama-French five-factor model (Panel B) along with ± 2 standard error bands around each estimate. Portfolios are constructed from the Fama-French 25 size and value portfolios based on a 20-day momentum signal. The figure reports performance of the simple factor (“Factor,” that is, the standard own-signal strategy, included as a benchmark), the equal-weighted average of the top three principal portfolios (“PP 1-3”), the equal-weighted average of the top three principal exposure portfolios (“PEP 1-3”), the equal-weighted average of the top three principal alpha portfolios (“PAP 1-3”), and the equal-weighted average of the top three PEPs and PAPs combined (“PEP and PAP 1-3”). Results are shown for forecast horizons (and, equivalently, holding periods) of 1, 5, 10, and 20 days. Each forecast is made on an out-of-sample basis using a rolling training sample of the 120 most recent nonoverlapping return observations. The sample period is 1963 to 2019.

Panel A: Sharpe Ratio



Panel B: Information Ratio



Panel C: Correlation between average portfolio returns and eigenvalues

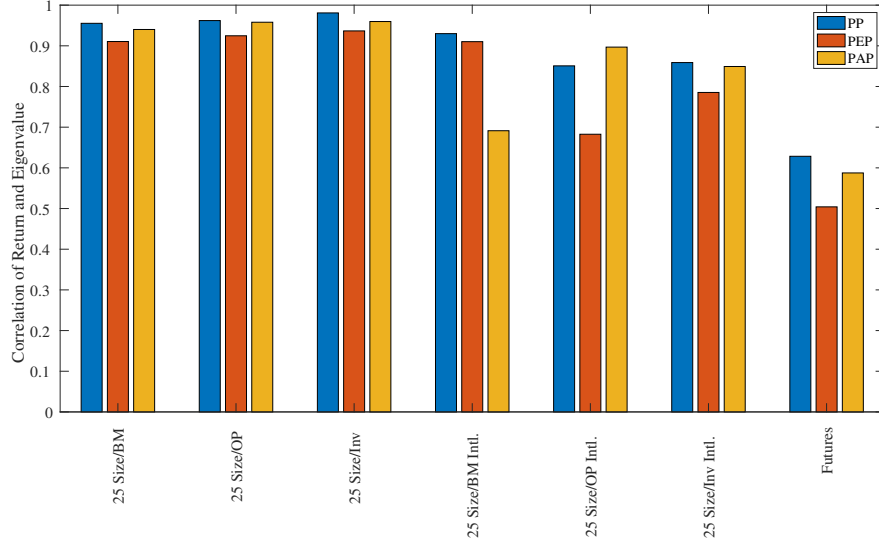


Figure IA2. Principal portfolio performance in other asset universes. This figure shows out-of-sample performance of principal portfolios in terms of the annualized Sharpe ratio (Panel A) and the annualized information ratio versus the own-predictor strategy and the Fama-French five-factor model (Panel B) along with ± 2 standard error bands around each estimate. Portfolios are constructed based on a 20-day momentum signal from either the 25 U.S. size and value portfolios, 25 U.S. size and operating profitability portfolios, 25 U.S. size and investment portfolios, their international counterparts (developed countries excluding the U.S.), or 52 futures contracts. The figure shows performance of the simple factor (“Factor,” that is, the standard own-signal strategy, included as a benchmark), the equal-weighted average of the top three principal portfolios (“PP 1-3”), the equal-weighted average of the top three principal exposure portfolios (“PEP 1-3”), the equal-weighted average of the top three principal alpha portfolios (“PAP 1-3”), and the equal-weighted average of the top three PEPs and PAPs combined (“PEP and PAP 1-3”). Panel C shows the correlation between out-of-sample average portfolio returns and eigenvalues of the prediction matrix. Blue bars show the correlation between principal portfolios and singular values from the total prediction matrix, red bars show the correlation between PEPs and eigenvalues from the symmetric component, and yellow bars show the correlation between PAPs and eigenvalues from the antisymmetric component. Results are shown for a 20-day forecast horizon/holding period. Each forecast is made on an out-of-sample basis using a rolling training sample of the 120 most recent nonoverlapping return observations. The sample period is 1963 to 2019 for U.S. equity portfolios, 1990 to 2019 for international equity portfolios, and 1985 to 2019 for futures contracts.

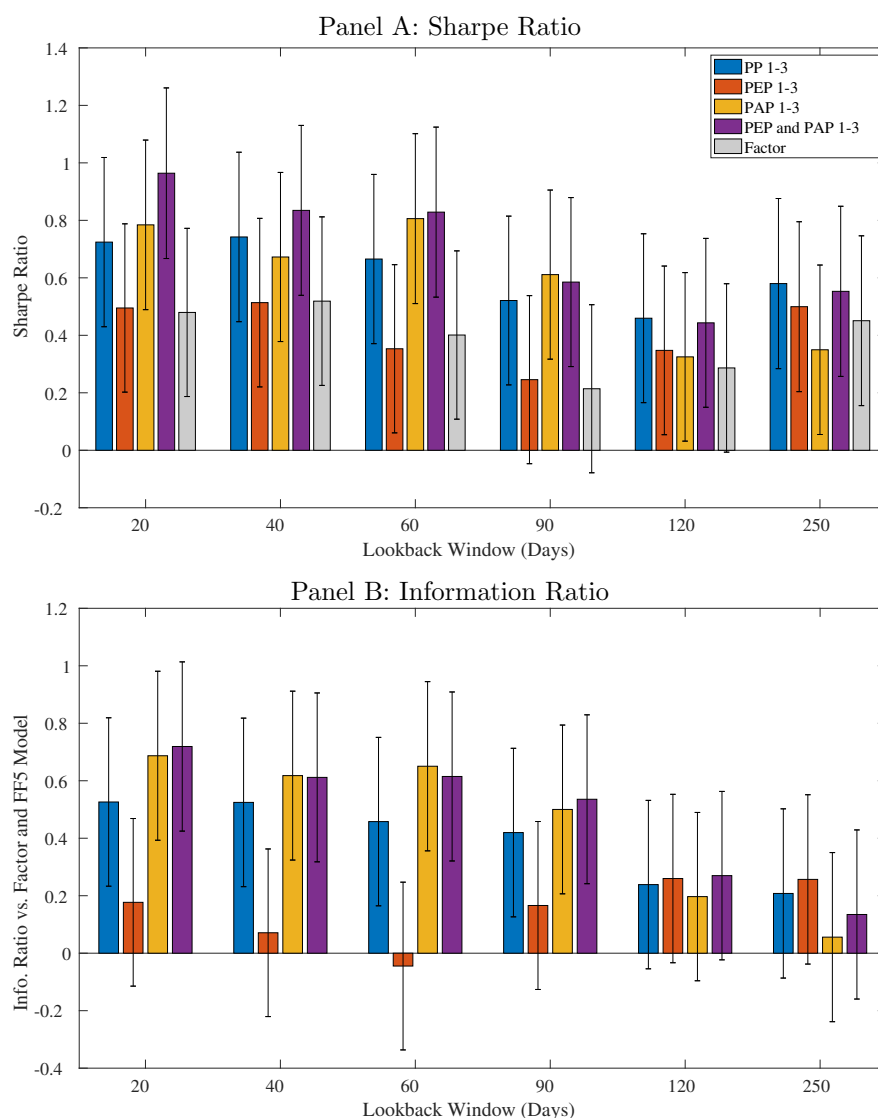


Figure IA3. Principal portfolio performance by momentum lookback window. This figure shows out-of-sample performance of principal portfolios in terms of the annualized Sharpe ratio (Panel A) and the annualized information ratio versus the own-predictor strategy and the Fama-French five-factor model (Panel B) along with ± 2 standard error band around each estimate. Portfolios are constructed based on a 20-, 40-, 60-, 120- or 250-day momentum signal from 25 size and value portfolios. The figure reports performance of the simple factor (“Factor”), the equal-weighted average of the top three principal portfolios (“PP 1-3”), the equal-weighted average of the top three principal exposure portfolios (“PEP 1-3”), the equal-weighted average of the top three principal alpha portfolios (“PAP 1-3”), and the equal-weighted average of the top three PEPs and PAPs combined (“PEP and PAP 1-3”). Each forecast is made on an out-of-sample basis using a rolling training sample of the 120 most recent nonoverlapping return observations. The sample period is 1963 to 2019.

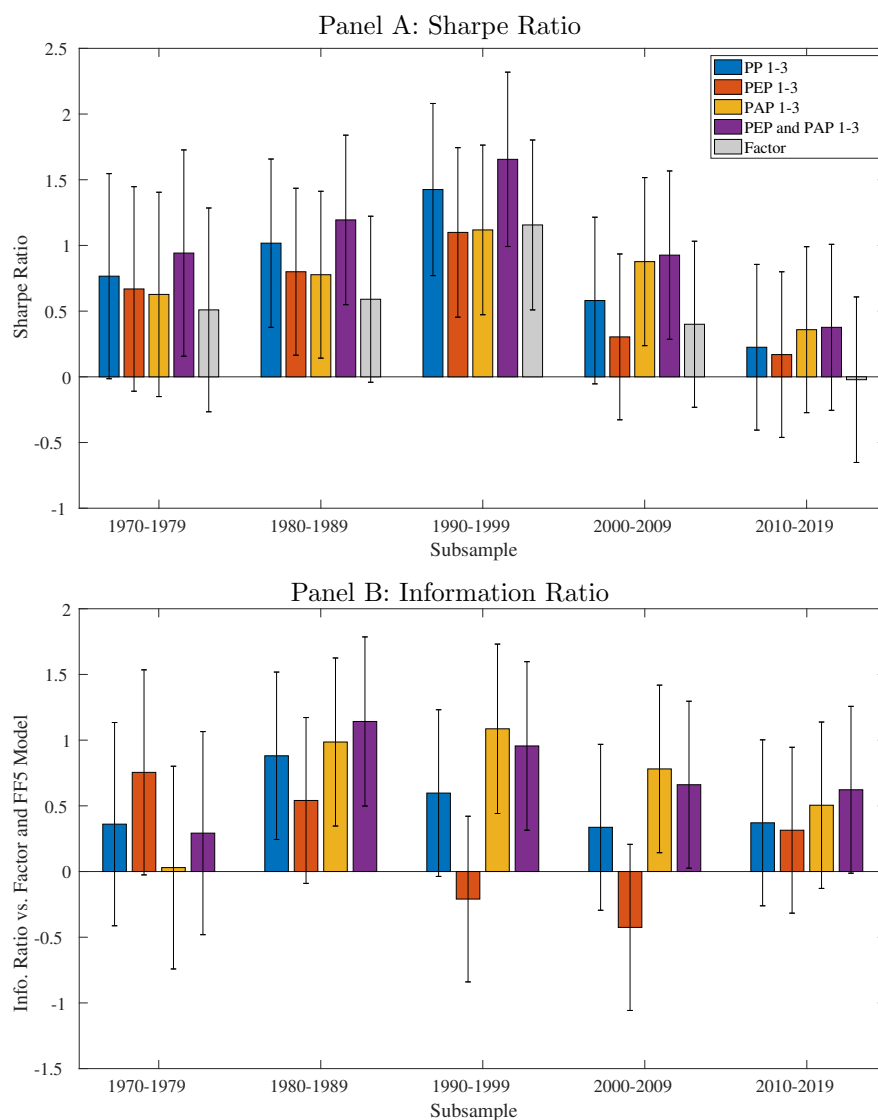


Figure IA4. Principal portfolio performance in subsamples. This figure shows out-of-sample performance of principal portfolios by decade in terms of the annualized Sharpe ratio (Panel A) and the annualized information ratio versus the own-predictor strategy and the Fama-French five-factor model (Panel B) along with ± 2 standard error band around each estimate. Portfolios are constructed based on a 20-day momentum signal from 25 size and value portfolios. The figure reports performance of the simple factor (“Factor”), the equal-weighted average of the top three principal portfolios (“PP 1-3”), the equal-weighted average of the top three principal exposure portfolios (“PEP 1-3”), the equal-weighted average of the top three principal alpha portfolios (“PAP 1-3”), and the equal-weighted average of the top three PEPs and PAPs combined (“PEP and PAP 1-3”). Results are shown for a 20-day forecast horizon/holding period. Each forecast is made on an out-of-sample basis using a rolling training sample of the 120 most recent nonoverlapping return observations.

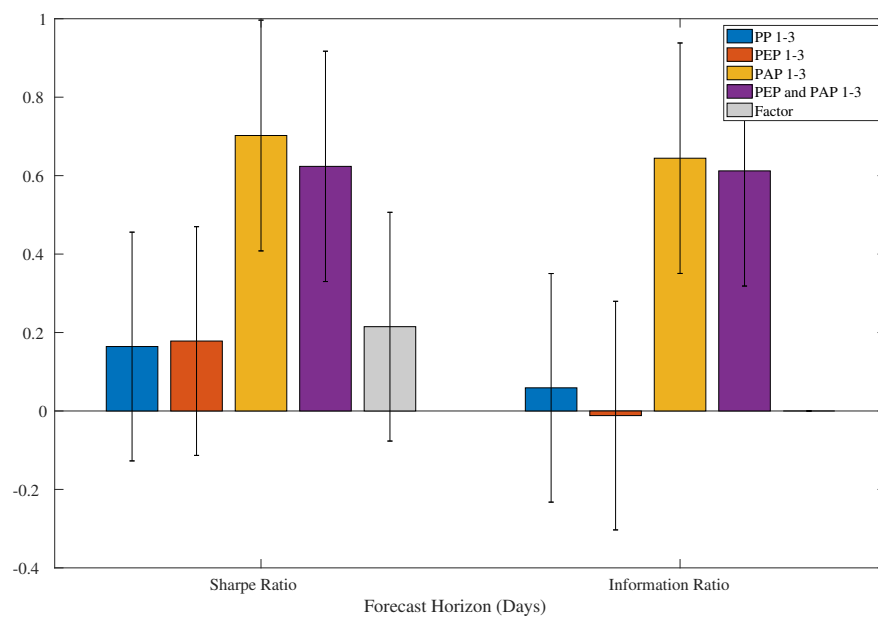


Figure IA5. Principal portfolio performance ratios with no centering. This figure repeats the analysis of Figure 3 but with uncentered signals and returns.

Appendix References

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